Infosys Springboard Virtual Internship 7.0

Internship Completion Report

Group Name : Group 1

Batch Number : Batch 01

Start Date : 29-06-2026

Team Details : Team A

Team Members :

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Internship Duration : 8 weeks

1.Project Title:

Supply Chain Visibility System with Optimization Analytics Group 1

2.Project Objective :

The main objective of this project is to analyze and visualize supply chain data to understand overall business, product, customer, delivery, market, order, discount, profit, and shipping performance. The project focuses on transforming raw supply chain data into meaningful business insights using Microsoft Power BI. By creating interactive dashboards and performing detailed analysis, the project helps understand the overall performance of supply chain operations and supports data-driven business decision-making.

The project aims to achieve the following objectives:

- Preprocess and clean the DataCo Supply Chain dataset by removing unnecessary columns, handling missing values and duplicates, correcting data types, replacing inconsistent values, and creating useful derived columns such as Year and Quantity Range using Power Query Editor.
- Analyze overall business performance by evaluating total sales, total orders, total profit, average shipping days, product quantity sold, average product price, and discount amount.
- Study product performance by identifying the top-selling products, product categories, departments, pricing patterns, and product-wise sales contribution to understand inventory and pricing performance.
- Analyze customer behavior through customer segments, customer states, customer cities, and average sales per customer to identify high-performing customer groups and geographical locations.
- Evaluate delivery and shipping performance by examining delivery status, shipping modes, scheduled shipping days, late deliveries, and shipping efficiency across different markets.
- Compare regional and market performance across Europe, LATAM, Pacific Asia, USCA, and Africa to identify the markets contributing the highest sales, profits, and customer orders.
- Analyze order and discount performance by studying order quantity levels, order status distribution, discount rates, discount trends, and their relationship with product sales.
- Measure profitability by evaluating profit ratio, benefit per order, discount amount, department-wise sales trends, and shipping-mode profitability.
- Develop seven interactive Power BI dashboards that present different perspectives of supply chain performance through KPI cards, charts, maps, treemaps, slicers, tables, decomposition trees, and other interactive visualizations.

- Identify important trends and patterns in sales growth, customer distribution, product demand, delivery efficiency, regional performance, shipping behavior, discounts, and profitability over multiple years.
- Generate meaningful business insights that help improve inventory management, pricing strategies, logistics planning, customer analysis, regional sales performance, and overall supply chain efficiency.
- Present the complete analysis through easy-to-understand and interactive dashboards so that business users can quickly monitor performance and make informed operational decisions.
- Demonstrate the practical application of Business Intelligence and Data Analytics concepts learned during the Infosys Springboard Virtual Internship by solving a real-world supply chain business problem.

Overall, this project aims to convert complex supply chain data into a structured and interactive business intelligence solution that enables organizations to monitor supply chain performance, identify operational challenges, and make better strategic decisions through data visualization and analytics.

3.Project Description in detail:

3.1 Overview

The Supply Chain Visibility and Optimization project is a Business Intelligence and Data Analytics project developed as part of the Infosys Springboard Virtual Internship 7.0. The project focuses on analyzing supply chain operations using the DataCo Supply Chain Dataset to understand different business activities, including orders, products, customers, sales, pricing, discounts, profits, shipping, delivery status, and market performance. The main purpose of the project is to convert raw supply chain data into meaningful insights that help organizations monitor performance, identify operational challenges, and improve overall supply chain efficiency through interactive dashboards.

The project began with collecting and understanding the dataset, followed by a complete data preprocessing and transformation process using Power Query Editor in Microsoft Power BI. During preprocessing, unnecessary columns were removed, missing values and duplicate records were handled, incorrect data types were corrected, inconsistent values were cleaned, and useful derived columns such as Year, Quantity Range, and other analytical fields were created. These preprocessing steps ensured that the dataset was accurate, clean, and suitable for visualization and business analysis.

After data preprocessing, the cleaned dataset was imported into Microsoft Power BI Desktop to build seven interactive dashboards. These dashboards include Supply Chain Performance Dashboard, Product Performance & Pricing Dashboard, Customer &

Geographic Analysis Dashboard, Delivery & Shipping Performance Dashboard, Regional & Market Performance Dashboard, Order & Discount Performance Dashboard, and Profitability & Order Insights Dashboard. Each dashboard focuses on a specific area of the supply chain and uses KPI cards, bar charts, line charts, pie charts, treemaps, maps, tables, slicers, and other visualizations to present information in an interactive and user-friendly manner.

The dashboards provide detailed insights into sales performance, product demand, customer distribution, delivery efficiency, regional sales trends, discount analysis, shipping performance, and profitability across different markets and customer segments. The project helps identify top-performing products and markets, understand customer purchasing patterns, evaluate logistics performance, monitor discount trends, and measure overall business profitability. By presenting these insights through interactive visualizations, the project supports data-driven decision-making, improves supply chain visibility, and demonstrates the practical application of Business Intelligence concepts in solving real-world supply chain problems.

3.2 Project Approach

The Supply Chain Visibility System with Optimization Analytics project followed a structured and systematic approach to transform raw supply chain data into interactive business intelligence dashboards. The project was executed through multiple phases, including data collection, data understanding, preprocessing, cleaning, visualization, dashboard development, and business insight generation. Each phase was completed sequentially to ensure that the dataset was accurate, well-organized, and suitable for analysis. The overall workflow helped convert complex supply chain information into meaningful visual reports for better decision-making.

Step 1: Problem Statement

- The first step was to identify the business problem and understand the importance of supply chain visibility. The objective was to analyze sales, customer behavior, product performance, shipping efficiency, delivery status, regional performance, discounts, and profitability using Power BI dashboards. A clear problem statement helped define the project scope, objectives, expected outputs, and business requirements before starting the analysis.

Step 2: Data Collection

- The DataCo Supply Chain Dataset was collected as the primary dataset for this project. The dataset contains detailed information related to customer orders, products, departments, categories, markets, shipping modes, delivery status, pricing, discounts, profits, and sales transactions. The dataset was imported into Microsoft

Power BI Desktop and verified to ensure that all records and columns were successfully loaded for further processing and analysis.

Step 3: Data Understanding

- Before preprocessing, the dataset was carefully studied to understand its structure and business meaning. All columns were examined to identify customer information, order details, product details, shipping information, market information, pricing details, discounts, and profitability fields. Relationships between different attributes were understood to determine which columns were useful for KPI calculations and dashboard visualizations. This step helped identify unnecessary columns and important analytical fields.

Step 4: Data Preprocessing

- Data preprocessing was performed using Power Query Editor to prepare the dataset for business analysis. During preprocessing, data types were converted into appropriate formats such as Date, Text, Whole Number, and Decimal Number. Customer names were split into first and last names, date columns were transformed into a standard format, and custom columns such as Order Year and Quantity Range were created. Additional transformations like merging columns, replacing values, filtering rows, and checking column quality were also performed to improve the dataset structure.

Step 5: Data Modelling

- After preprocessing, data modelling techniques were applied to improve data quality. Unnecessary columns with empty or irrelevant values were removed, duplicate records were identified and deleted, missing values were handled, and inconsistent records were corrected wherever required. Invalid records were filtered, and the final dataset was verified to ensure that it was clean, accurate, and analysis-ready. This step improved the reliability of all dashboard calculations and visualizations.

Step 6: Data Visualization

- The cleaned dataset was used to create multiple visualizations in Microsoft Power BI. Different visualization techniques were selected based on business requirements. KPI Cards were created for important business metrics, while Bar Charts, Column Charts, Line Charts, Pie Charts, Donut Charts, Treemaps, Scatter Charts, Maps, Funnel Charts, Waterfall Charts, Ribbon Charts, Tables, Matrices, and Slicers were used to analyze different aspects of supply chain performance. Interactive visuals helped users filter and explore business data easily.

Step 7: Dashboard Development

Using the visualizations created in Power BI, seven interactive dashboards were developed, each focusing on a different area of supply chain analysis.

- Supply Chain Performance Dashboard – Overall business performance, sales, orders, profit, and delivery status.
- Product Performance & Pricing Dashboard – Product sales, categories, departments, pricing, and top-selling products.
- Customer & Geographic Analysis Dashboard – Customer segments, customer locations, states, cities, and geographic sales distribution.
- Delivery & Shipping Performance Dashboard – Shipping modes, delivery status, late deliveries, shipping days, and logistics performance.
- Regional & Market Performance Dashboard – Market-wise sales, customer segments, regional performance, and sales target comparison.
- Order & Discount Performance Dashboard – Order quantity, discounts, order status, quantity levels, and sales decomposition analysis.
- Profitability & Order Insights Dashboard – Profit ratio, total discounts, departmental sales trends, shipping profitability, and order insights.

Each dashboard includes interactive filters, slicers, and multiple visualizations to provide detailed business analysis.

Step 8: Insights & Analysis

- The final step involved analyzing the dashboards and generating meaningful business insights. Sales performance was compared across markets and product categories, customer purchasing behavior was analyzed using customer segments and locations, delivery performance was evaluated through shipping status and shipping modes, and profitability was studied using discount trends and department-wise sales. The generated insights helped identify high-performing markets, profitable products, customer trends, logistics challenges, and opportunities for improving supply chain operations. These findings were documented in the internship report and presented through interactive dashboards to support data-driven business decision-making.

3.3 Tools and Technologies Used :

The following tools and technologies were used throughout the Supply Chain Visibility System with Optimization Analytics project for data preprocessing, dashboard development, collaboration, documentation, and project submission.

Tools	Purpose
DataCo Supply Chain Dataset (CSV)	Used as the primary dataset for supply chain analysis and imported into Power BI for visualization and reporting.
Microsoft Power BI Desktop	Used for creating interactive dashboards, KPI cards, charts, maps, slicers, tables, and business intelligence reports.
Power Query Editor	Used for data cleaning, preprocessing, data transformation, handling missing values, changing data types, filtering rows, splitting and merging columns, and creating custom columns.
DAX (Data Analysis Expressions)	Used to create calculated measures and KPIs such as Total Sales, Total Profit, Total Orders, Average Shipping Days, Average Product Price, and Sales Target.
Microsoft Excel	Used to view, verify, and organize the dataset before importing it into Power BI and for converting the dataset into CSV format.
GitHub	Used to upload and maintain the complete project repository, including README documentation, dashboard screenshots, preprocessing screenshots, PPTs, reports, and dataset files.
GitLab	Used as an additional project repository for storing and sharing project files, maintaining version control, and submitting internship project deliverables.
Microsoft Teams	Used for team collaboration, project meetings, mentor discussions, milestone reviews, presentation sessions, and communication throughout the internship.
Microsoft Word	Used to prepare the internship completion report, project documentation, objectives, methodology, results, and conclusions.
Microsoft PowerPoint	Used to create both the Individual PPT and Team PPT for project presentation and milestone demonstrations.

3.4 Insights from the Dashboard :

3.4a Supply Chain Performance Dashboard Insights :

1. Total Orders (KPI Card – 180.519K)

- The dashboard shows that the company processed 180.519K customer orders, indicating a large-scale supply chain operation.

- A high number of orders reflects strong customer demand and continuous business transactions across different markets.
- This KPI helps management understand the overall order volume handled during the analysis period.

2. Total Profit (KPI Card – 3.97M)

- The company generated an overall profit of approximately 3.97 Million, showing positive business performance.
- Total profit helps evaluate how efficiently the company converts sales into earnings.
- This KPI is useful for monitoring overall financial performance and profitability.

3. Total Sales (KPI Card – 36.78M)

- The company achieved total sales of approximately 36.78 Million across all markets and product categories.
- This KPI provides a complete overview of revenue generated during the selected period.
- It helps compare business performance across different dashboards and reporting periods.

4. Average Actual Shipping Days (KPI Card – 3.50 Days)

- The average actual shipping time was 3.50 days, indicating the overall delivery efficiency of the supply chain.
- This metric helps evaluate whether customer orders are delivered within the expected time frame.
- Monitoring average shipping days helps identify opportunities to improve logistics and delivery performance.

5. Delivery Status Analysis (Donut Chart)

- Late Delivery accounts for the highest percentage (54.83%) of all deliveries, making it the most common delivery status.
- Advance Shipping and Shipping on Time together represent a significant share of successful deliveries.
- A smaller percentage of Shipping Cancelled orders indicates that only a limited number of deliveries were cancelled.

6. Sales by Market (Bar Chart)

- Europe recorded the highest sales (10.9 Million), making it the best-performing market.
- LATAM and Pacific Asia also contributed a large share of total sales, while Africa generated the lowest sales.
- Market-wise sales comparison helps identify regions with stronger business growth and revenue potential.

7. Sales Trend Over Time (Line Chart)

- Sales remained almost stable during 2015, 2016, and 2017, with values around 12 Million.
- A sharp decline is observed in 2018, where sales reduced to nearly 0.3 Million.
- The yearly sales trend helps monitor business growth, identify performance changes over time, and support future sales forecasting.

3.4b Product Performance & Pricing Dashboard Insights :

1. Total Products (KPI Card – 180.519K)

- The dashboard shows a total of 180.519K product records available in the supply chain dataset.
- A large number of products indicates a wide product portfolio managed by the company across different categories and departments.
- This KPI provides an overview of the total products involved in supply chain operations.

2. Average Product Price (KPI Card – 141.23)

- The average selling price of products is approximately 141.23.
- This KPI helps understand the overall pricing level of products across the business.
- Average product price is useful for comparing pricing strategies and identifying premium or low-cost product segments.

3. Top 10 Products by Sales (Bar Chart)

- The Field & Stream Sportsman product generated the highest sales of approximately 6.9 Million.
- Other products such as Perfect Fitness, Diamondback, and Nike Men's products also contributed significantly to total sales.
- Identifying top-selling products helps the company prioritize inventory planning and product promotion strategies.

4. Sales by Category (Column Chart)

- Women's Apparel and Water Sports categories recorded the highest sales among all product categories.
- Categories such as Toys, Video Games, and Women's Clothing contributed comparatively lower sales.
- Category-wise sales analysis helps identify customer demand and high-performing product categories.

5. Sales by Department (Treemap)

- The Fan Shop department generated the highest sales, contributing approximately 17.11 Million.
- Apparel, Golf, and Footwear departments also contributed a significant share of overall sales.
- Department-wise analysis helps identify business units that generate maximum revenue and profitability.

6. Product Price and Sales (Scatter Chart)

- The scatter chart shows the relationship between product price and sales performance for different products.
- Products with medium price ranges generated higher sales compared to some expensive products.
- This visualization helps understand customer purchasing patterns and the impact of product pricing on sales.

3.4c Customer & Geographic Analysis Dashboard Insights :

1. Total States (KPI Card – 46 States)

- The company serves customers across 46 different states, showing a wide geographical presence.
- A large customer base spread across multiple states indicates strong regional market coverage.
- This KPI helps understand the overall geographic reach of the supply chain network.

2. Average Sales per Customer (KPI Card – 203.77)

- The average sales generated per customer is 203.77.
- This metric helps evaluate the average purchasing value of customers across all regions.

- It is useful for understanding customer spending behavior and planning customer engagement strategies.

3. Customer Distribution by State (Filled Map)

- The filled map shows customer distribution across different states in North America.
- Some states have a higher concentration of customers, indicating stronger market penetration in those regions.
- Geographic visualization helps identify regions with higher customer activity and business opportunities.

4. Sales by Customer City (Bar Chart)

- York recorded the highest customer sales among all cities included in the dashboard.
- Cities such as Ypsilanti, Wyoming, Yonkers, and Wyandotte also contributed significant sales.
- City-wise sales analysis helps identify high-performing customer locations and target markets for future growth.

5. Top 10 States by Sales (Column Chart)

- Puerto Rico (PR) generated the highest sales, contributing approximately 14.2 Million.
- States such as Texas (TX), Virginia (VA), and Tennessee (TN) contributed comparatively lower sales than Puerto Rico.
- This visualization helps compare sales performance across the top-performing states.

6. Customer Segment Distribution (Donut Chart)

- The Consumer segment contributes the largest share of customers, accounting for approximately 51.8% of the total customer base.
- Corporate customers contribute around 30%, while Home Office customers represent about 18%.
- Segment-wise distribution helps the company understand which customer group contributes most to business sales.

7. Profit Contribution by Product (Treemap)

- The Web Camera product contributes the highest profit among the products displayed in the dashboard.

- Other products such as Under Armour items also contribute a noticeable share of profit.
- Product-wise profit analysis helps identify the most profitable products and supports better product and inventory planning.

3.4d Delivery & Shipping Performance Dashboard Insights :

1. Total Late Deliveries (KPI Card – 99K)

- The dashboard shows approximately 99K late deliveries, indicating that delayed shipments form a significant portion of total deliveries.
- A high number of late deliveries highlights the need to improve logistics and delivery planning.
- This KPI helps monitor delivery performance and customer service efficiency.

2. Average Scheduled Shipping Days (KPI Card – 2.93 Days)

- The average scheduled shipping time is 2.93 days across all orders.
- This metric represents the planned delivery duration before shipment.
- Comparing scheduled shipping days with actual delivery performance helps identify delivery delays.

3. Year Filter (Slicer)

- The dashboard allows users to analyze delivery performance for 2015, 2016, 2017, and 2018 individually.
- Year-wise filtering helps compare shipping and delivery trends across different years.
- It provides flexibility to study changes in logistics performance over time.

4. Delivery Status by Shipping Mode (100% Stacked Column Chart)

- First Class deliveries have the highest percentage of on-time shipping compared to other shipping modes.
- Standard Class shows the highest percentage of late deliveries among all shipping methods.
- This visualization helps compare delivery efficiency across different shipping modes.

5. Profit Contribution by Market (Waterfall Chart)

- Europe contributes the highest share of profit among all markets.
- LATAM, Pacific Asia, USCA, and Africa add additional profit contributions to the overall total.

- The waterfall chart clearly shows how each market contributes to cumulative business profit.

6. Shipping Performance Trend by Year (Ribbon/Stacked Trend Chart)

- Shipping performance remains strong during 2015–2017, with higher sales across major markets.
- Europe and LATAM consistently contribute significant shipping-related sales.
- The trend helps monitor market-wise logistics performance over different years.

3.4e Regional & Market Performance Dashboard Insights :

1. Average Benefits per Order (KPI Card – 21.97)

- The average benefit earned per order is 21.97.
- This KPI measures the profitability generated from each customer order.
- It helps evaluate overall business returns across markets.

2. Total Markets (KPI Card – 5 Markets)

- The business operates across five major markets: Europe, LATAM, Pacific Asia, USCA, and Africa.
- This KPI provides an overview of the company's regional business presence.
- It helps compare performance across different markets.

3. Total Sales Achievement (Gauge Chart)

- Total sales reached approximately 36.78 Million, achieving the defined business sales target.
- The gauge chart compares actual sales against the maximum target value.
- It helps management evaluate overall sales performance quickly.

4. Average Sales by Market (Column Chart)

- Europe recorded the highest average sales among all markets.
- Pacific Asia and LATAM also maintained strong average sales performance.
- Africa and USCA generated comparatively lower average sales.

5. Customer Segment Performance (Table)

- The Consumer segment contributed the highest sales, profit, and order count.
- Corporate customers generated the second-highest contribution.

- Home Office customers contributed the smallest share among the three customer segments.

6. Market Wise Sales Summary (Table)

- Europe generated the highest total sales and profit contribution.
- LATAM ranked second in both sales and order volume.
- Africa recorded the lowest sales and order count among the five markets.

7. Orders by Shipping Mode (Pie Chart)

- Standard Class handled the highest percentage of customer orders.
- Second Class and First Class contributed moderate order volumes.
- Same Day shipping represented the smallest share of total orders.

3.4f Order & Discount Performance Dashboard Insights :

1. Total Quantity Sold (KPI Card – 384K)

- The company sold approximately 384K product quantities during the analysis period.
- This KPI represents the overall order quantity handled by the supply chain.
- Higher quantity sold indicates strong customer demand.

2. Average Discount Rate (KPI Card – 18.35K)

- The dashboard displays the overall discount value applied across order items.
- Discounts play an important role in influencing customer purchasing behavior.
- This KPI helps monitor promotional and pricing strategies.

3. Orders by Quantity Level (Funnel Chart)

- Most customer orders fall within the 1–2 quantity range, contributing the highest order volume.
- Orders containing 3–5 products occur less frequently.
- This analysis helps understand customer purchasing quantity patterns.

4. Order Status Distribution by Market (Bar Chart)

- Completed orders represent the largest share across all markets.
- Europe records the highest number of completed orders compared to other markets.

- Different markets show variations in pending, closed, cancelled, and payment-related orders.

5. Discount Rate vs Sales by Product (Scatter Chart)

- Products with higher discount rates do not always generate the highest sales.
- Some moderately discounted products achieve better sales performance than heavily discounted products.
- The scatter chart helps analyze the relationship between discount strategy and product sales.

6. Sales Breakdown Analysis (Decomposition Tree)

- Total sales can be analyzed by Market → Category → Shipping Mode.
- Europe contributes the highest market sales before breaking down into product categories.
- The decomposition tree provides a detailed drill-down analysis of sales performance.

3.4g Profitability & Order Insights Dashboard :

1. Average Profit Ratio (KPI Card – 0.12)

- The average profit ratio is 0.12, representing the profitability earned from order items.
- This KPI helps evaluate the profit margin across supply chain operations.
- It provides a quick overview of business profitability.

2. Total Discount Amount (KPI Card – 3.73M)

- Customers received discounts worth approximately 3.73 Million.
- This KPI measures the total value of discounts offered across all orders.
- It helps evaluate the financial impact of discount campaigns.

3. Average Benefit by Shipping Mode (Line Chart)

- Same Day shipping generated the highest average benefit during 2018.
- First Class and Standard Class showed higher benefits during 2017.
- Shipping mode profitability changes across different years.

4. Sales by Department over Years (Stacked Area Chart)

- Golf department contributed the highest sales during 2015–2017.

- Outdoor and Technology departments also contributed noticeable sales.
- Department-wise yearly analysis helps identify long-term business growth.

5. Discount Trend (Area Chart)

- Discount values remained around 1.2–1.3 Million during 2015–2017.
- A significant decline is observed during 2018.
- The discount trend helps monitor promotional activity over time.

6. Order Status Distribution (Pie Chart)

- Complete orders account for the largest percentage of total orders.
- Pending Payment, Processing, and Closed orders contribute moderate shares.
- Cancelled, Suspicious Fraud, and Payment Review orders represent a smaller percentage.

3.5 Real-World Impact for Media and Public Communication :

The Supply Chain Visibility and Optimization project demonstrates how Data Visualization and Business Intelligence can transform complex supply chain information into meaningful and easy-to-understand business insights. Organizations generate large amounts of data related to sales, orders, products, customers, shipping, delivery status, discounts, and profitability every day. Through interactive Power BI dashboards, this project converts raw operational data into visual reports that help users quickly understand business performance without analyzing large datasets manually. The dashboards provide a clear overview of key supply chain activities and enable users to communicate important business information in a simple and effective way.

The project has a significant impact on organizational communication and business reporting because it provides a common platform for different departments to access the same business information. Sales teams can monitor revenue and customer trends, logistics teams can track shipping and delivery performance, inventory teams can identify product demand, and management can evaluate profitability and market performance. By presenting business metrics through KPI cards, charts, maps, tables, and slicers, the dashboards improve communication between departments and ensure that decisions are based on accurate and real-time information.

The interactive nature of the dashboards also supports media and public communication by presenting business information in a visually attractive and understandable format. Users can filter the dashboards by year, market, shipping mode, customer segment, product category, and delivery status to generate customized reports for different audiences. These visual reports make it easier to explain business performance during presentations, review meetings, project demonstrations, and stakeholder discussions. The dashboards simplify complex supply chain data into clear visual stories that can be understood by both technical and non-technical users.

Overall, the project demonstrates the practical value of Power BI-based reporting in improving transparency, accountability, and decision-making within supply chain management. It helps organizations identify performance gaps, compare regional and market performance, monitor operational activities, evaluate delivery efficiency, and recognize opportunities for improvement. The insights generated through this project can support strategic planning, customer satisfaction, logistics optimization, and business growth while promoting effective communication through interactive dashboards, reports, and presentations.

4.Timeline Overview :

Week	Activities Planned	Activities Completed
Week1	• Project understanding and objective discussion. • Collect the DataCo Supply Chain Dataset. • Understand dataset structure, columns, and business problem. • Plan data preprocessing workflow.	Understood the supply chain problem statement. • Collected and imported the DataCo Supply Chain Dataset. • Identified important columns for sales, orders, customers, products, shipping, and profit analysis. • Prepared the dataset for preprocessing in Power BI.
Week2	• Perform data cleaning using Power Query Editor. • Remove missing values and duplicate records. • Correct data types and replace incorrect values. • Create derived columns required for analysis.	Removed unnecessary columns and handled blank values. • Checked duplicates and corrected inconsistent records. • Converted date columns into proper date format. • Created Year, Quantity Range and other calculated processing columns.
Week3	• Complete data preprocessing and transformation. • Load cleaned dataset into Power BI. • Create DAX measures and KPI calculations.	• Completed feature selection and data transformation. • Created KPIs such as Total Sales, Total Profit, Total Orders, and Average Shipping Days.

	• Begin dashboard development.	• Connected the cleaned dataset to Power BI. • Started designing the first dashboards.
Week4	• Develop Supply Chain Performance Dashboard. • Create Product Performance & Pricing Dashboard. • Build Customer & Geographic Analysis Dashboard. • Add interactive visuals and filters.	• Developed three interactive Power BI dashboards. • Added KPI cards, bar charts, donut charts, treemaps, line charts, and maps. • Included slicers for better dashboard interaction.
Week5	• Develop Delivery & Shipping Dashboard. • Develop Regional & Market Performance Dashboard. • Create Order & Discount Performance Dashboard. • Analyze logistics and market trends.	• Completed three additional dashboards. • Analyzed delivery status, shipping modes, regional sales, discounts, and order performance. • Generated business insights from each visualization.
Week6	• Develop Profitability & Order Insights Dashboard. • Review dashboard accuracy and consistency. • Validate KPIs and business insights.	• Completed the seventh dashboard successfully. • Verified all KPI calculations and DAX measures. • Reviewed charts, slicers, and filters for consistency.

Week7	• Analyze dashboard results and prepare insights. • Prepare project report and documentation. • Create team and individual presentations.	• Analyzed delivery, transportation, inventory, supplier, customer, and operational performance. • Prepared dashboard explanations and project findings. • Completed the internship report, team PPT, and individual PPT.
Week8	• Finalize project documentation. • Organize project files. • Upload project to GitHub and GitLab. • Prepare final project demonstration and presentation.	• Completed dashboard analysis and documentation. • Organized Power BI dashboard, screenshots, report, dataset, and presentation files. • Uploaded the project repository to GitHub and GitLab. • Prepared the final project demonstration and presentation for evaluation.

5a. Key Milestones :

Milestone	Description	Date Achieved
Project Kickoff	Understood the supply chain problem statement, formed the team, collected the DataCo Supply Chain dataset, finalized the project objectives, and planned the complete workflow for the internship project.	02/07/2026
Prototype / First Draft	Completed data preprocessing and data cleaning using Power Query Editor, corrected data types, created derived columns, and developed the initial version of the Power BI dashboard with KPIs and basic visualizations.	16/07/2026
Mid-Term Review	Reviewed dashboard visuals, refined KPI calculations, improved dashboard layout, added interactive slicers and charts, and incorporated mentor feedback to enhance dashboard performance and presentation.	29/07/2026

Final Submission	Finalized all Power BI dashboards, completed the project documentation, validated the dashboard insights and business analysis, organized project files, and submitted the completed internship project successfully.	19/08/2026
Presentation	Presented the completed project, explained the project methodology, preprocessing steps, dashboard insights, and demonstrated all interactive Power BI dashboards during the final review.	20/08/2026

Milestone 1 – Data Preprocessing and Dataset Preparation

The first milestone focused on collecting, understanding, preprocessing, and preparing the DataCo Supply Chain Dataset for business analysis. Initially, the dataset was collected in CSV format and carefully examined to understand its structure, number of records, columns, and the different types of supply chain information available. The dataset contained information related to customer details, product categories, departments, order information, sales, discounts, shipping modes, delivery status, markets, regions, profits, and pricing. Understanding the dataset helped identify which attributes were important for dashboard development and business analysis.

After understanding the dataset, the preprocessing stage was performed using Power Query Editor in Microsoft Power BI. During this process, unnecessary columns such as empty and irrelevant fields were removed, duplicate records were checked and eliminated, missing values and errors were handled, and column names were renamed for better readability. The data types of date, text, decimal, and integer columns were corrected to ensure accurate calculations and visualizations. This improved the consistency and quality of the dataset before analysis.

Several transformation techniques were applied during preprocessing to make the dataset analysis-ready. Customer names were split into first and last names, columns were merged where required, filtered rows were applied, and custom columns such as Order Year and Quantity Range were created for better reporting. Date columns were converted into the correct format using locale settings, and column quality was verified before loading the dataset into Power BI. These preprocessing steps ensured that the dataset became clean, organized, and suitable for dashboard creation.

At the end of Milestone 1, the cleaned dataset was successfully loaded into Power BI Desktop and prepared for visualization. This milestone established the foundation for the remaining project by ensuring that the data was accurate, reliable, and ready for business intelligence reporting.

Milestone 2 – Development of First Two Dashboards

The second milestone focused on developing the first set of interactive dashboards using the cleaned dataset in Microsoft Power BI. This phase introduced the Business Intelligence layer of the project by creating dashboards that summarized overall supply chain performance and product-related analysis. DAX measures were created for important KPIs such as Total Sales, Total Profit, Total Orders, Average Shipping Days, and Average Product Price.

The Supply Chain Performance Dashboard was developed to provide an executive overview of the business. KPI cards displayed Total Orders, Total Sales, Total Profit, and Average Shipping Days, while charts analyzed sales by market, delivery status distribution, and yearly sales trends. The dashboard highlighted Europe as the highest-performing market and identified late deliveries as an important logistics challenge. Interactive slicers allowed users to analyze data for different years and markets.

The Product Performance & Pricing Dashboard was created to understand product sales and pricing behavior. Visualizations analyzed top-selling products, sales by product category, department-wise sales, product pricing, and the relationship between product price and sales using scatter charts. This dashboard helped identify high-performing products, profitable categories, and departments contributing the highest revenue.

By completing this milestone, the project successfully produced the first two dashboards with interactive filters, KPI cards, and business visuals. These dashboards provided valuable insights into overall business performance and product-level analysis, forming the core reporting layer of the project.

Milestone 3 – Development of Customer, Delivery and Regional Dashboards

The third milestone expanded the project by developing three additional dashboards that focused on customer analysis, delivery performance, and regional market performance. The objective of this milestone was to analyze supply chain operations from customer, logistics, and geographical perspectives using interactive Power BI visualizations.

The Customer & Geographic Analysis Dashboard analyzed customer distribution across states, cities, markets, and customer segments. KPI cards displayed the total number of states and average sales per customer, while maps and charts highlighted customer concentration and top-performing locations. The dashboard also identified Consumer as the largest customer segment and Puerto Rico as one of the highest-performing regions in terms of sales.

The Delivery & Shipping Performance Dashboard focused on logistics efficiency by analyzing delivery status, shipping modes, scheduled shipping days, late deliveries, and market-wise profitability. Charts compared delivery performance across different shipping methods and years, helping identify areas where logistics operations could be improved. The dashboard also showed that late deliveries accounted for a significant percentage of overall deliveries.

The Regional & Market Performance Dashboard compared business performance across Europe, LATAM, Pacific Asia, USCA, and Africa. It included KPI cards, gauge charts for sales target achievement, customer segment analysis, market-wise sales comparison, and shipping mode distribution. This milestone provided a comprehensive understanding of regional business growth and customer contribution across multiple markets.

Milestone 4 – Development of Order, Discount and Profitability Dashboards

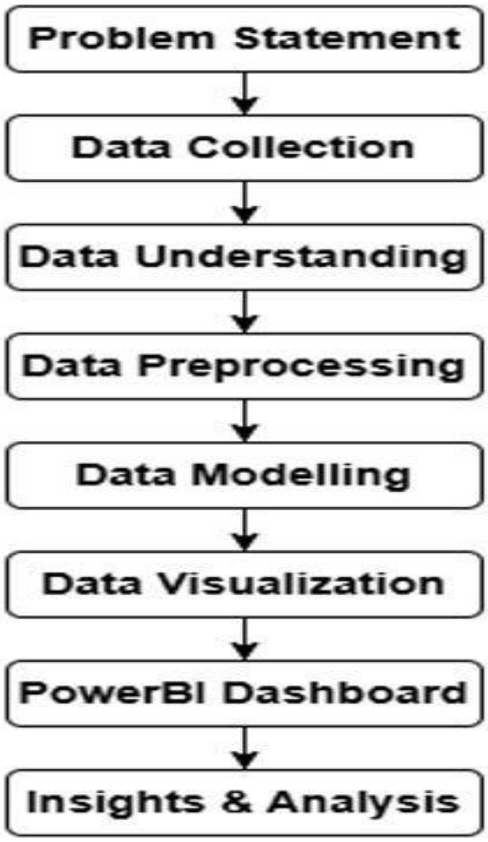
The fourth milestone completed the remaining business analysis by developing the final two dashboards: Order & Discount Performance Dashboard and Profitability & Order Insights Dashboard. These dashboards focused on customer order behavior, discount strategies, profitability analysis, and financial performance.

The Order & Discount Performance Dashboard analyzed order quantity levels, quantity ranges, discount rates, order status distribution, sales decomposition, and discount impact on product sales. Funnel charts, scatter charts, decomposition trees, and bar charts helped identify customer purchasing behavior, discount patterns, and sales contribution across markets and shipping modes. This dashboard supported better pricing and promotional decision-making.

The Profitability & Order Insights Dashboard focused on profit ratio, total discount amount, shipping mode profitability, department-wise sales trends, yearly discount trends, and order status analysis. KPI cards summarized overall profitability, while charts highlighted profitable departments and shipping methods. This dashboard helped understand the financial impact of discounts and shipping strategies on business performance.

After completing the final dashboards, all seven interactive Power BI dashboards were reviewed, validated, and optimized to ensure accuracy and consistency. Business insights were generated from each dashboard, documentation was completed, and the dashboards were finalized for report preparation, presentation, GitHub repository upload, GitLab submission, and internship project demonstration.

5b. Project Execution Details



Step 1: Problem Statement

The project was initiated to improve **supply chain visibility and business performance analysis** using Microsoft Power BI. The main objective was to transform raw supply chain data into meaningful and interactive visual insights that could help understand different areas of business operations. The analysis focused on **sales, orders, products, customers, pricing, discounts, profit, markets, shipping performance, delivery status, and regional performance**.

The project was designed to provide different business perspectives through multiple dashboards rather than combining all information into a single report. This approach makes it easier for management and users to monitor specific areas, identify performance gaps, compare business trends, and support **data-driven decision-making and operational improvement**.

Step 2: Data Collection

The **DataCo Supply Chain Dataset** was collected in CSV format from Kaggle. The dataset contains a large number of records covering different aspects of supply chain operations, including **customer information, product details, order transactions, sales, discounts, profit, shipping, delivery status, markets, regions, categories, and departments**.

The collected dataset was initially examined to understand the available fields and their relevance to the project requirements. The CSV dataset was then imported into **Microsoft**

Power BI, where Power Query Editor was used for preprocessing, transformation, and data cleaning before developing the dashboards.

Step 3: Data Understanding

The dataset was carefully studied before beginning the visualization process. The available columns were examined based on their **business meaning, data type, relationship with other fields, and usefulness for dashboard analysis**.
Important fields related to **Order ID, Customer ID, Customer Full Name, Customer Segment, Customer City, Customer State, Product Name, Product Price, Category Name, Department Name, Sales, Order Item Quantity, Discount, Profit Ratio, Benefit per Order, Market, Order Region, Shipping Mode, Delivery Status, Late Delivery Risk, Order Status, Order Date, and Shipping Date** were identified.
The data was divided conceptually into areas such as **order analysis, product analysis, customer analysis, delivery and shipping analysis, market and regional analysis, discount analysis, and profitability analysis**. This understanding helped in selecting appropriate columns and visualizations for the seven dashboards.
Step 4: Data Preprocessing
Data preprocessing was performed using **Power Query Editor** to convert the raw dataset into a structured and analysis-ready format. Multiple transformation steps were applied to improve the consistency, readability, and usability of the dataset.

Major preprocessing activities included:

- **Changed Data Types:** Corrected the data types of date, text, integer, decimal, and other numerical columns according to their actual values.
- **Changed Type with Locale:** Used appropriate locale settings for date/time fields to correctly interpret date values.
- **Removed Unnecessary Columns:** Removed columns that were not required for the final seven dashboards and analysis.
- **Removed Duplicate Records:** Checked and removed duplicate records wherever applicable to improve data consistency.
- **Merged Columns:** Combined relevant fields where required to create meaningful information for analysis.
- **Renamed Columns:** Changed column names into clearer and more readable names for easier identification in Power BI.
- **Replaced Values:** Corrected or standardized inconsistent values using the Replace Values transformation.
- **Removed Errors:** Identified and removed error records generated during data transformation.
- **Filtered Rows:** Applied row-level filters wherever invalid or unnecessary records were identified.

- **Split Customer Name:** The Customer Full Name field was transformed into separate customer name components for better customer-level analysis.
- **Added Custom Columns:** Created additional columns required for analysis and dashboard development.
- **Inserted Year:** Extracted the year from the order date to support year-wise sales and performance analysis.
- **Added Conditional Column:** Created the **Quantity Range** classification to group orders according to their order quantity.
- **Sorted Rows:** Sorted relevant data to improve organization and analysis.
- **Final Column Review:** After dashboard development, the dataset was reviewed again and unnecessary columns were removed to maintain a cleaner dataset.

These preprocessing steps ensured that the data was **clean, consistent, readable, and suitable for creating accurate Power BI visualizations**.

Step 5: Data Modelling

Data cleaning was performed after and along with preprocessing to improve the overall quality of the dataset. The objective was to reduce data quality problems that could affect calculations, KPIs, charts, and dashboard insights.
The following activities were carried out:

- Removed unnecessary and unused columns.
- Removed duplicate records where identified.
- Handled blank and missing values.
- Removed records containing transformation errors.
- Corrected incorrect or inconsistent data types.
- Standardized column names for better readability.
- Replaced inconsistent values where required.
- Filtered invalid or unwanted records.
- Checked numerical and date fields for proper formatting.
- Verified important sales, order, customer, product, and shipping fields.
- Reviewed the cleaned dataset before connecting it to the dashboard visuals.
- Performed a final validation to ensure that the dashboard calculations were based on properly prepared data.

After cleaning, the transformed dataset was connected to Power BI for analysis and visualization.

Step 6: Data Visualization

After preprocessing and cleaning, different Power BI visualizations were selected according to the type of business information being analyzed. The objective was to make complex supply chain information easier to understand and compare.
The following visualizations were used across the seven dashboards:

- **KPI Cards** – Used to display important numerical indicators such as total sales, total orders, profit/benefit, quantity, and average values.
- **Bar Charts** – Used to compare products, categories, departments, markets, states, and other business dimensions.
- **Line Charts** – Used to analyze yearly trends such as sales, shipping performance, benefits, and discounts.
- **Pie Charts** – Used to show the distribution of orders across shipping modes.
- **Donut Charts** – Used to analyze delivery status, customer segments, and order-status distribution.
- **Treemaps** – Used to show product, department, and profitability contributions.
- **Scatter Charts** – Used to analyze relationships such as product price vs sales and discount rate vs sales.
- **Maps/Filled Maps** – Used for geographical customer and state-level analysis.
- **Tables** – Used to provide detailed comparisons of markets and customer segments.
- **Waterfall Chart** – Used to analyze market-wise contribution to overall benefit.
- **Decomposition Tree** – Used to break down sales by market, category, and shipping mode.
- **Funnel Chart** – Used to analyze orders according to quantity levels.
- **Gauge Chart** – Used to compare actual sales against the defined sales target.
- **Slicers** – Used to allow interactive filtering, particularly for year-based analysis.

Step 7: Dashboard Development

After completing the data preparation and visualization process, **seven interactive Power BI dashboards** were developed. Each dashboard was designed for a different business perspective and contains relevant KPIs and visualizations.

Dashboard 1 – Supply Chain Performance Dashboard

This dashboard provides an overall view of supply chain performance.

- Displays **Total Orders, Total Benefit/Profit, Total Sales, and Average Actual Shipping Days** through KPI cards.
- Analyzes overall **Delivery Status** using a donut chart.

- Compares **Sales by Market** to identify high-performing markets.
- Displays **Sales Trend Over Time** to analyze yearly changes in sales.
- Helps management understand overall business performance and delivery efficiency.

Dashboard 2 – Product Performance & Pricing Dashboard

This dashboard focuses on product-level sales, pricing, categories, and departments.

- Displays the **Total Products** and **Average Product Price** through KPI cards.
- Identifies **Top 10 Products by Sales**.
- Compares **Sales by Product Category**.
- Analyzes **Sales by Department** using a treemap.
- Uses a **Product Price vs Sales Scatter Chart** to understand the relationship between product pricing and sales.
- Helps identify high-performing products and supports product pricing and inventory-related decisions.

Dashboard 3 – Customer & Geographic Analysis Dashboard

This dashboard analyzes customer distribution and customer-related sales performance.

- Displays **Total States** and **Average Sales per Customer** through KPI cards.
- Uses a geographical map to show **Customer Distribution by State**.
- Compares **Sales by Customer City**.
- Identifies the **Top 10 States by Sales**.
- Analyzes **Customer Segment Distribution** using a donut chart.
- Uses a treemap to analyze **Profit Contribution by Product**.
- Helps understand customer concentration, customer segments, and geographical sales patterns.

Dashboard 4 – Delivery & Shipping Performance Dashboard

This dashboard focuses on delivery efficiency and shipping performance.

- Displays **Total Late Deliveries** and **Average Scheduled Shipping Days**.
- Provides a **Year Slicer** for year-wise analysis.
- Compares **Delivery Status by Shipping Mode** using a 100% stacked chart.
- Analyzes **Profit/Benefit Contribution by Market** using a waterfall chart.
- Displays **Shipping Performance Trend by Year**.

- Helps identify delivery issues, compare shipping modes, and understand market-wise logistics performance.

Dashboard 5 – Regional & Market Performance Dashboard

This dashboard focuses on regional, market, customer segment, and shipping performance.

- Displays **Average Benefit per Order** and **Total Markets** using KPI cards.
- Uses a **Gauge Chart** to compare actual sales with the predefined sales target.
- Compares **Average Sales by Market**.
- Provides a detailed **Customer Segment Performance** table containing sales, benefit, and order information.
- Provides a **Market-wise Sales Summary** for comparing different markets.
- Uses a pie chart to analyze **Orders by Shipping Mode**.
- Helps identify strong markets, customer segments, sales achievement, and order distribution.

Dashboard 6 – Order & Discount Performance Dashboard

This dashboard focuses on order quantity, discount behavior, and sales analysis.

- Displays **Total Quantity Sold** and **Average/Discount Rate information** through KPI cards.
- Uses a **Funnel Chart** to analyze **Orders by Quantity Level**.
- Compares **Order Status Distribution by Market**.
- Uses a scatter chart to analyze the relationship between **Discount Rate and Sales by Product**.
- Uses a **Decomposition Tree** to break down sales through different dimensions such as market, category, and shipping mode.
- Helps understand customer order quantity patterns, discount behavior, and factors contributing to sales.

Dashboard 7 – Profitability & Order Insights Dashboard

This dashboard focuses on profitability, discounts, shipping modes, departments, and order status.

- Displays **Average Profit Ratio** and **Total Discount Amount** through KPI cards.
- Compares **Average Benefit by Shipping Mode across Years**.
- Analyzes **Sales by Department over Years**.

- Displays the **Discount Trend over Time**.
- Uses a donut chart to analyze **overall Order Status Distribution**.
- Helps understand profitability, discount impact, departmental sales trends, and the overall order lifecycle.

Step 8: Insights & Business Recommendations

After completing the seven dashboards, the analyzed data provided several important business insights.

Key Insights

- More than **180K orders** were processed, indicating a large volume of supply chain activity.
- **Europe** emerged as one of the strongest markets in terms of sales performance.
- **Standard Class** accounted for the largest share of orders among shipping modes.
- Late deliveries represent an important area for logistics improvement.
- The **Consumer customer segment** contributed a major share of customer activity.
- Product and category analysis helped identify important sales contributors.
- Product pricing and sales showed different patterns across products.
- Sales and performance varied across different markets, states, cities, categories, and departments.
- Discount analysis helped identify the relationship between discount levels and product sales.
- Year-wise analysis helped identify changes in sales, shipping performance, department performance, and discount trends.
- Profitability analysis helped compare the contribution of different products, markets, departments, and shipping modes.

Business Recommendations

- Improve logistics and delivery planning to reduce late deliveries.
- Monitor shipping modes with higher delivery issues and improve their operational efficiency.
- Maintain sufficient inventory for high-performing products and categories.
- Focus marketing and sales strategies on high-performing markets and customer segments.
- Regularly monitor low-performing markets, products, and departments.

- Review discount strategies to ensure that increased discounts contribute effectively to sales.
- Use product pricing and sales analysis to support better pricing decisions.
- Monitor profitability across shipping modes, products, and markets.
- Use the interactive Power BI dashboards for continuous performance monitoring and **data-driven strategic decision-making**.

6. Snapshots/Screenshots :

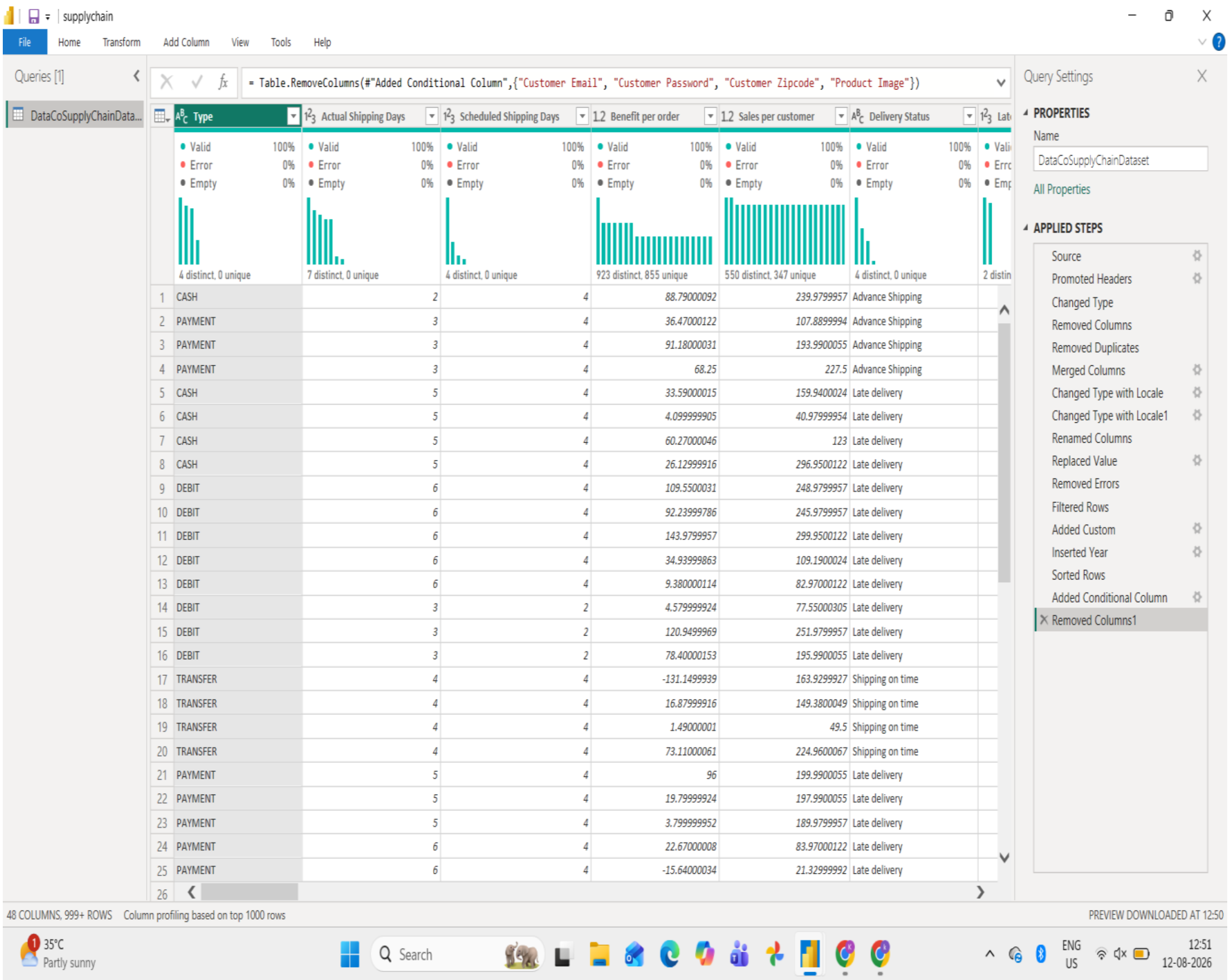


fig 1 : Data Preprocessing

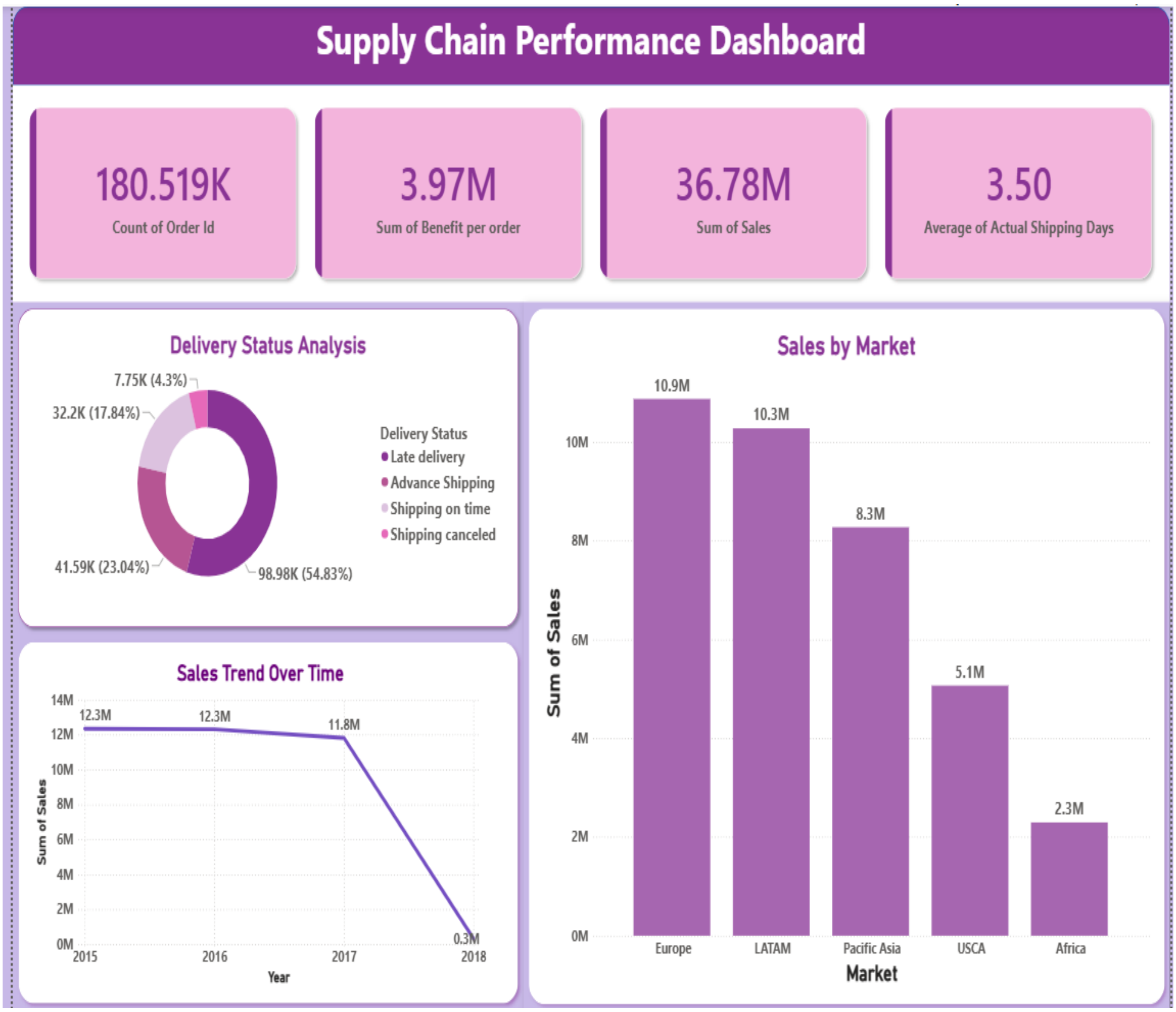


fig 2 : Supply Chain Performance Dashboard

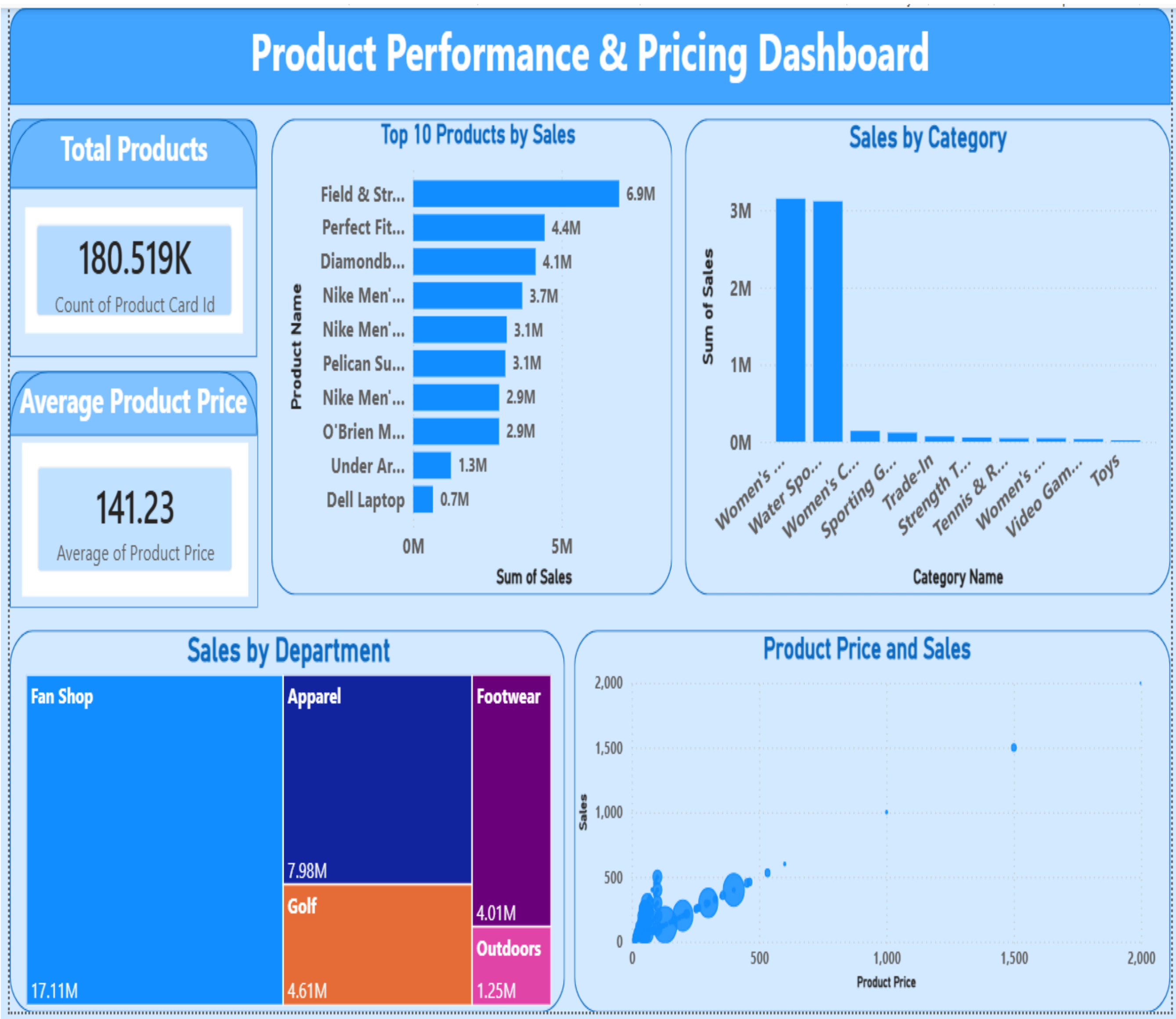


fig 3 : Product Performance & Pricing Dashboard

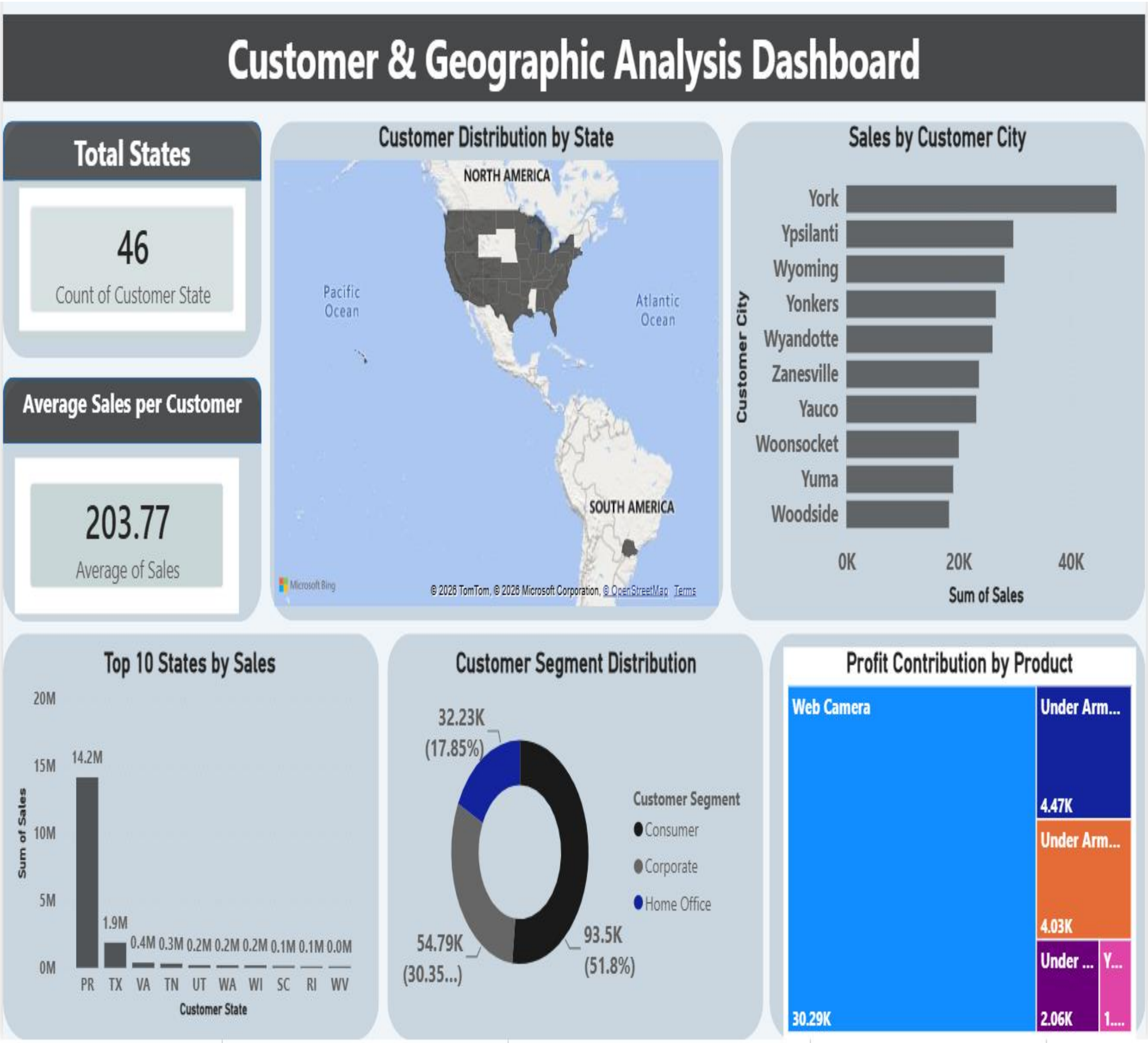


fig 4 : Customer & Geographic Analysis Dashboard

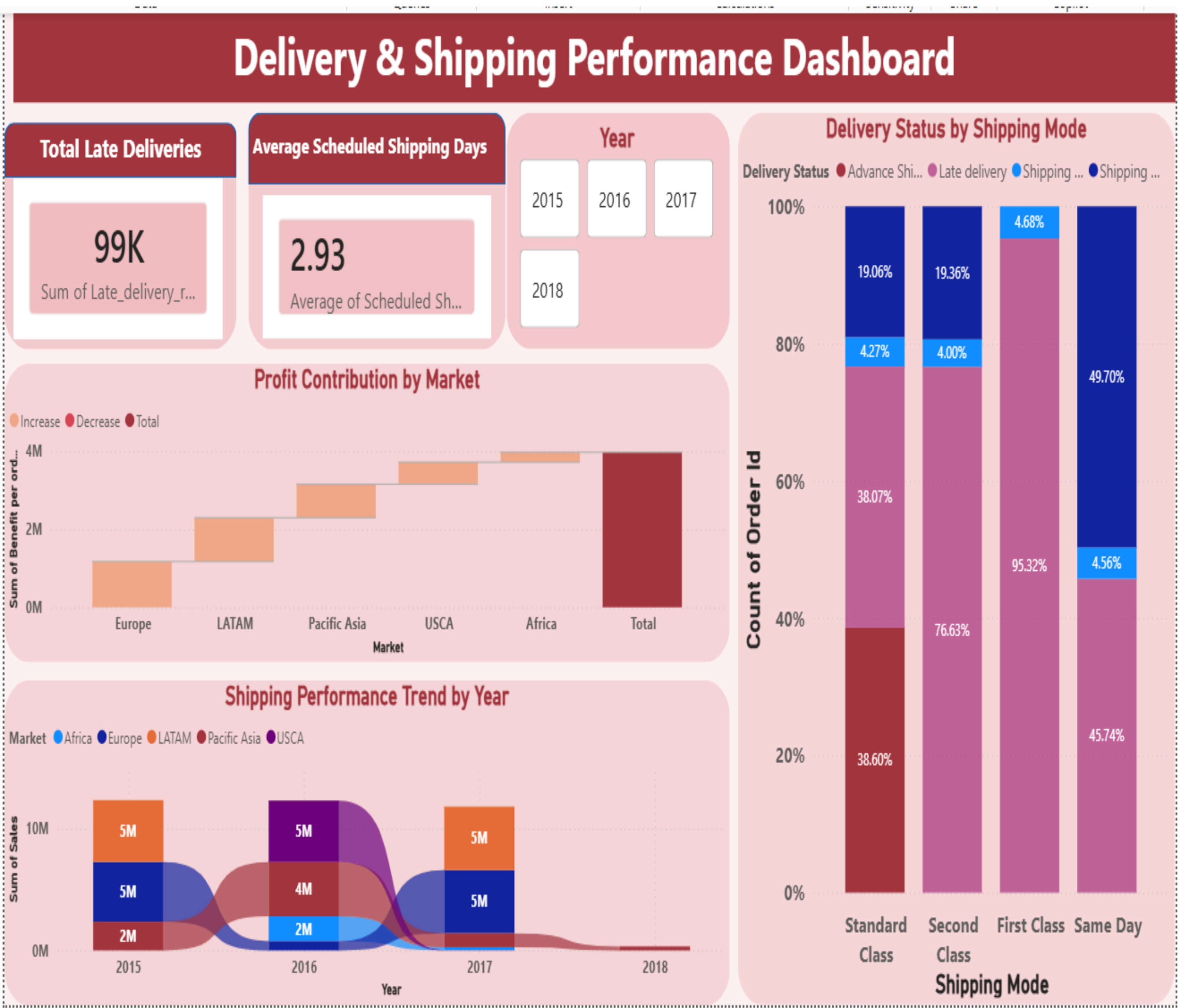


fig 5 : Delivery & Shipping Performance Dashboard

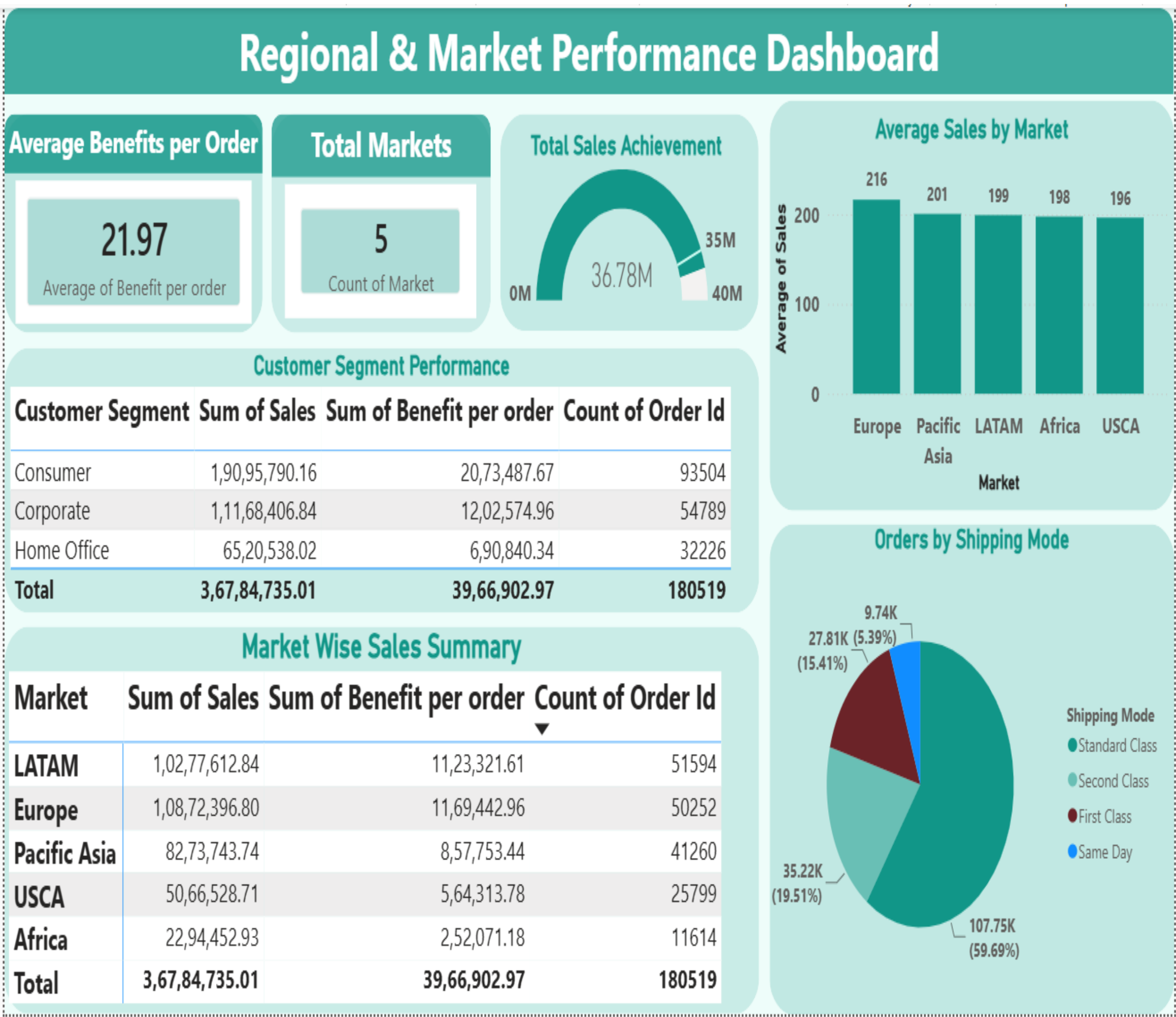


fig 6 : Regional & Market Performance Dashboard

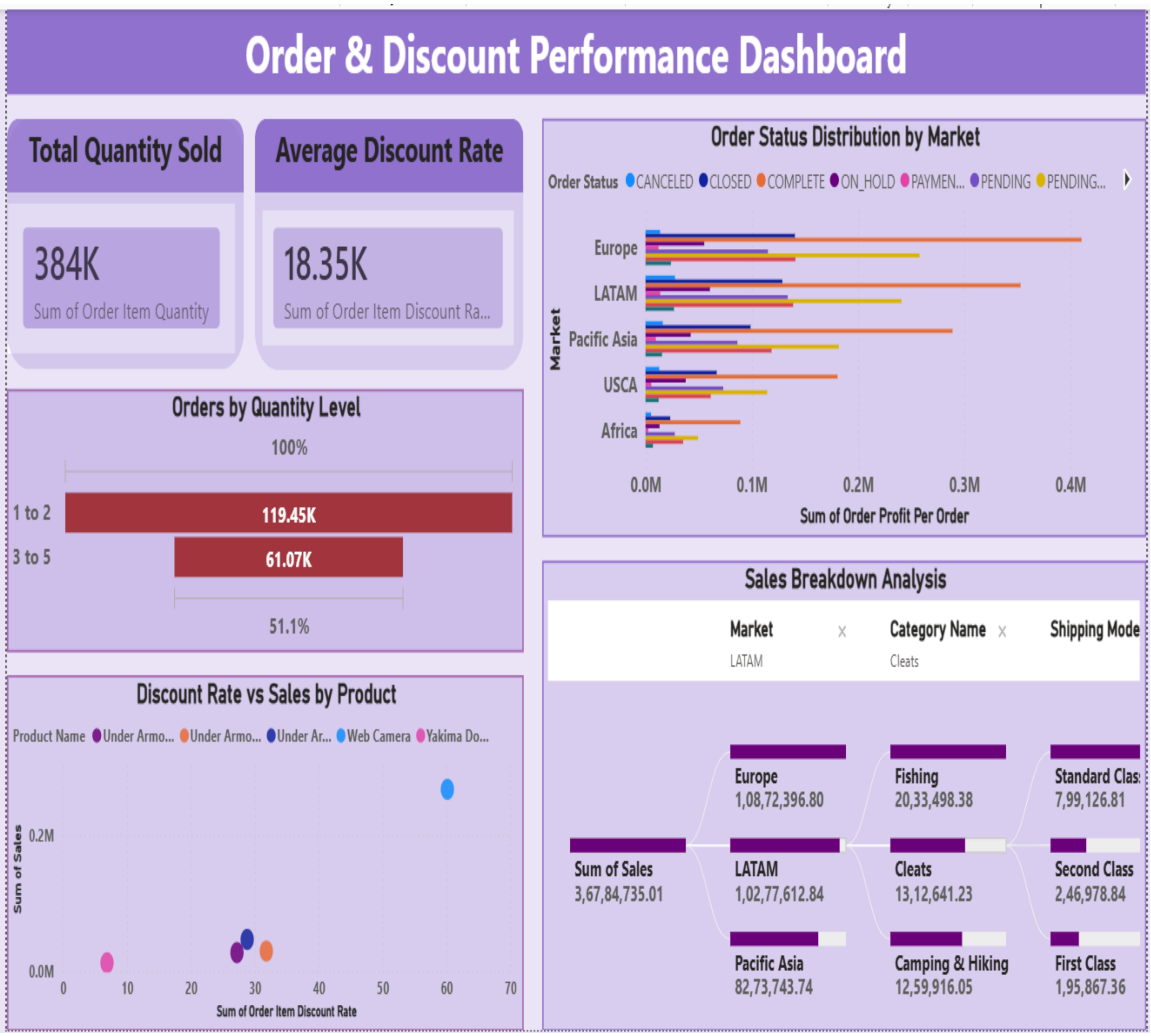


fig 7 : Order & Discount Performance Dashboard

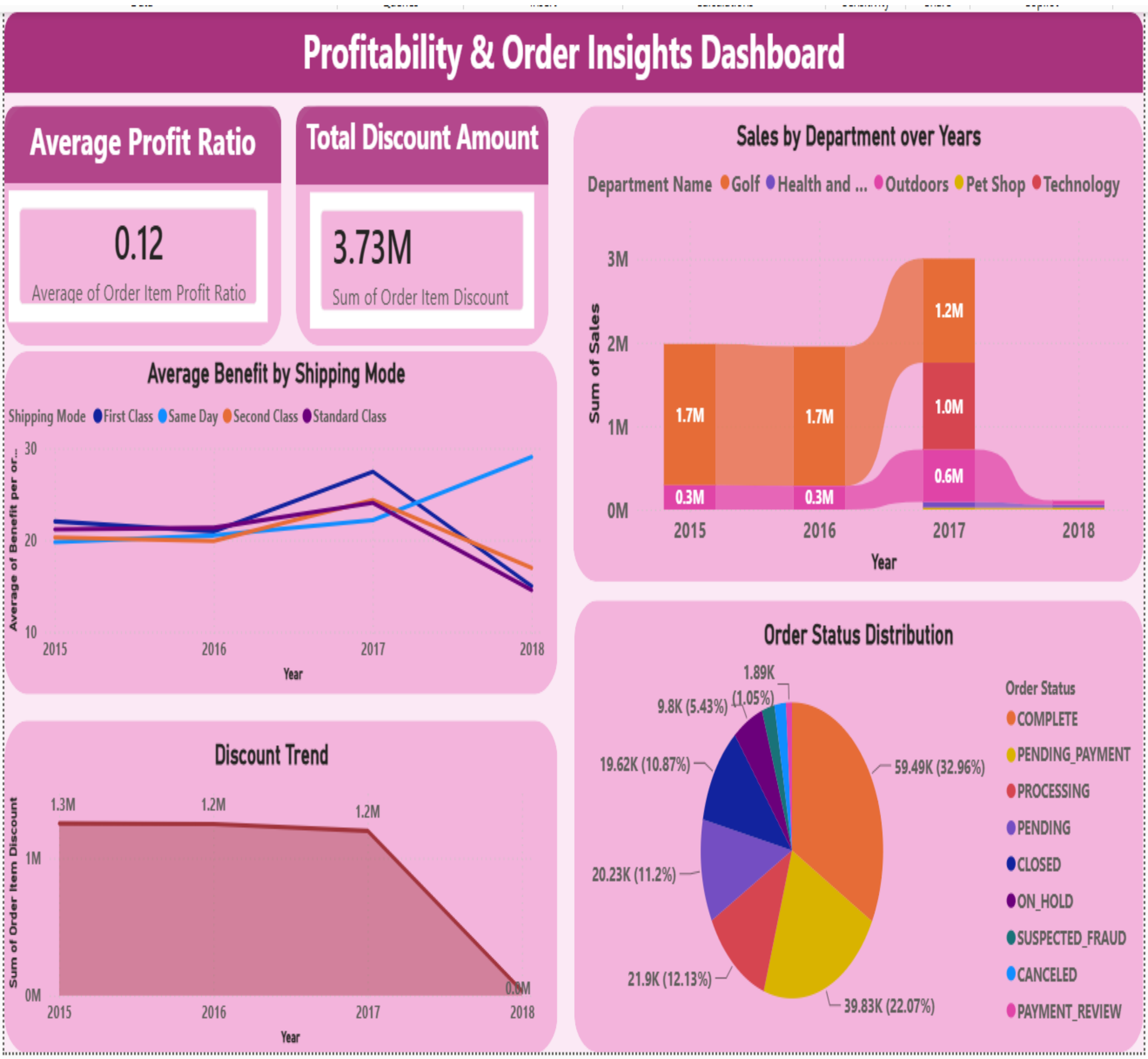


fig 8 : Profitability & Order Insights Dashboard

7. Challenges Faced :

1. Understanding the Dataset :

One of the major challenges was understanding the large DataCo Supply Chain Dataset, which contains numerous attributes related to customers, products, orders, shipping, sales, and markets. Identifying the relevant columns and understanding the relationships between different data fields required careful analysis before starting the dashboard development process.

2. Data Preprocessing and Cleaning :

The dataset contained missing values, duplicate records, inconsistent data types, and unnecessary columns. Cleaning and transforming the dataset using **Power Query Editor** required significant effort to ensure data accuracy and consistency. Proper preprocessing was essential for generating reliable dashboard results.

3. Selecting Appropriate Visualizations :

Choosing the right Power BI visualizations for different business metrics was another challenge. Different chart types such as KPI cards, bar charts, line charts, donut charts, treemaps, scatter plots, maps, and waterfall charts were carefully selected to present information clearly and effectively according to business requirements.

4. Designing Professional and Interactive Dashboards :

Developing visually appealing and user-friendly dashboards required careful planning of colors, layout, spacing, alignment, and formatting. It was also important to avoid repeating the same visualizations across different dashboards while maintaining a consistent professional design throughout the project.

5. Generating Business Insights :

Analyzing the dashboard results and extracting meaningful business insights was one of the most important challenges. Sales trends, customer behavior, product performance, shipping efficiency, and regional business performance were carefully interpreted to provide valuable recommendations that support better business decision-making.

8. Learnings & Skills Acquired :

8.1 Technical Skills :

1. Microsoft Power BI Dashboard Development :

Gained practical experience in using Microsoft Power BI to design and develop interactive business intelligence dashboards. Learned how to create professional reports using various visualizations, KPI cards, filters, slicers, and dashboard navigation to present business information effectively.

2. Data Cleaning and Transformation :

Developed hands-on experience in Power Query Editor for performing data preprocessing tasks such as correcting data types, handling missing values, removing duplicate records, filtering invalid data, renaming columns, splitting customer names, and transforming raw data into an analysis-ready format.

3. Interactive Data Visualization :

Improved technical skills in creating different Power BI visualizations, including KPI Cards, Bar Charts, Line Charts, Pie Charts, Donut Charts, Treemaps, Scatter Charts, Filled Maps, Waterfall Charts, Ribbon Charts, Tables, Matrix, Gauge Charts, and Slicers. Learned how to select appropriate visualizations based on different business requirements.

4. Business Reporting and Dashboard Design :

Enhanced knowledge of designing professional, user-friendly, and interactive dashboards by applying proper layouts, color themes, formatting, alignment, and visual consistency. Learned to present complex business data in a simple and understandable format for management and stakeholders.

5. Business Insight Generation :

Developed the ability to analyze supply chain data and generate meaningful business insights related to sales performance, customer behavior, product performance, delivery status, shipping efficiency, and regional performance. These insights support data-driven decision-making and improve overall business operations.

6. Analytical and Problem-Solving Skills :

Strengthened analytical thinking and problem-solving abilities by identifying data quality issues, selecting appropriate visualization techniques, interpreting dashboard results, and providing business recommendations based on data analysis.

8.2 Analytical and Problem-Solving Skills :

1. Data Analysis :

Developed the ability to analyze large DataCo Supply Chain datasets and identify meaningful business trends related to sales, profit, customer behavior, product performance, shipping efficiency, and regional business performance. Learned to interpret complex business data and convert it into valuable insights.

2. Data Quality Management :

Strengthened problem-solving skills by identifying and resolving data quality issues such as missing values, duplicate records, inconsistent data types, and invalid data. Applied appropriate data cleaning and preprocessing techniques using Power Query Editor to ensure accurate and reliable analysis.

3. Visualization and Analytical Thinking :

Improved analytical thinking by selecting the most appropriate Power BI visualizations for different business requirements. Evaluated various chart types to ensure that business information was presented in a clear, meaningful, and easy-to-understand format.

4. Business Decision Support :

Developed the ability to interpret dashboard results and generate meaningful business insights. Analyzed sales trends, customer segments, delivery performance, product performance, and regional sales to support effective business planning and data-driven decision-making.

5. Problem-Solving Approach :

Enhanced problem-solving skills by addressing real-world business challenges during dashboard development. Successfully identified operational issues, optimized dashboard design, improved report usability, and provided practical recommendations based on analytical findings.

6. Critical Thinking and Continuous Improvement :

Strengthened critical thinking by continuously reviewing dashboard outputs, validating results, and refining visualizations to improve accuracy and presentation quality. This process helped in delivering professional dashboards with reliable business insights.

8.3 Future Enhancements :

1. Real-Time Data Integration :

In the future, the dashboard can be enhanced by integrating real-time supply chain data from enterprise systems such as ERP or cloud databases. This will enable live monitoring of sales, inventory, orders, and delivery performance, allowing organizations to make faster and more informed business decisions.

2. Machine Learning-Based Demand Forecasting :

Advanced machine learning algorithms can be incorporated to predict future product demand, sales trends, and inventory requirements. This will help organizations optimize inventory planning, reduce stock shortages, minimize excess inventory, and improve overall supply chain efficiency.

3. Advanced Dashboard Features :

The dashboards can be further enhanced by adding advanced Power BI features such as drill-through reports, bookmarks, dynamic tooltips, advanced filters, and interactive navigation. These features will allow users to perform detailed analysis and explore business data more effectively.

4. Automated Data Refresh and Reporting :

Future improvements can include automated data refresh and scheduled report generation. This will ensure that dashboards always display the latest business information without requiring manual updates, thereby improving reporting efficiency and reducing operational effort.

5. Enhanced Business Analysis :

The dashboard can be expanded by including additional analytical modules such as inventory management, supplier performance analysis, warehouse operations, logistics optimization, and customer satisfaction analysis. These enhancements will provide a more comprehensive view of supply chain operations and support strategic business planning.

6. Scalability and Business Intelligence :

Future versions of the project can be designed to support larger datasets, multiple business locations, and cloud-based deployment. This will improve scalability, enable organization-wide reporting, and provide a centralized Business Intelligence platform for continuous performance monitoring and decision-making.

8.4 Soft Skills and Team Collaboration :

1. Communication and Presentation Skills :

Throughout the internship, communication and presentation skills were significantly improved by preparing project documentation, creating professional presentations, and explaining dashboard results clearly. This experience enhanced the ability to present technical information in a simple and understandable manner to different audiences.

2. Time Management :

Strong time management skills were developed by completing each project milestone according to the planned schedule. Tasks such as data preprocessing, dashboard development, documentation, and presentation preparation were effectively organized and completed within the given deadlines.

3. Problem-Solving and Critical Thinking :

The project provided opportunities to solve real-world business and technical challenges related to data quality, dashboard design, and business analysis. These activities strengthened analytical thinking, critical thinking, and the ability to identify practical solutions based on data.

4. Documentation and Project Organization :

Improved the ability to prepare well-structured project reports, maintain organized documentation, and present project progress in a professional manner. Proper documentation ensured that every stage of the project was clearly recorded and easy to understand.

5. Collaboration and Project Planning:

Enhanced collaboration and project planning skills by following a structured development approach throughout the internship. Regular review of project milestones, continuous improvements, and systematic planning helped in successfully completing the project and achieving the internship objectives.

6. Professional Development :

The internship helped build confidence, responsibility, adaptability, and a professional work approach. It also improved the ability to manage tasks independently, learn new technologies quickly, and apply analytical skills to solve business problems effectively.

8.5 Domain Knowledge and Application :

1. Supply Chain Management Knowledge :

The internship provided practical exposure to Supply Chain Management concepts, including sales analysis, order management, customer analysis, product performance, shipping operations, and delivery status monitoring. This helped in understanding how different supply chain activities are connected and contribute to overall business performance.

2. Business Performance Analysis :

Developed the ability to analyze key business metrics such as sales, profit, customer behavior, product categories, market performance, and delivery efficiency. Learned how these metrics can be used to evaluate organizational performance and identify areas for improvement.

3. Application of Business Intelligence :

Applied Business Intelligence (BI) techniques using Microsoft Power BI to transform raw supply chain data into meaningful visual reports. Interactive dashboards were created to simplify complex business information and support effective business analysis.

4. Decision-Making and Business Insights :

Gained an understanding of how dashboard visualizations support both operational and strategic decision-making. The generated insights help management monitor business performance, identify trends, improve supply chain efficiency, and make informed business decisions.

5. Real-World Business Application :

Applied theoretical concepts to a real-world supply chain dataset by performing data preprocessing, visualization, dashboard development, and business analysis. This practical experience improved the ability to solve business problems using data-driven approaches and Business Intelligence tools.

6. Team Collaboration :

Worked collaboratively throughout the internship by following project milestones, organizing project tasks, reviewing progress, and contributing to documentation, dashboard development, and presentation activities. This experience strengthened teamwork, coordination, and professional collaboration skills.

9.Testimonials from team :

Team Member : Kapu Tejaswini

The Supply Chain Visibility & Optimization internship project provided me with valuable hands-on experience in Data Analytics, Business Intelligence, and Supply Chain Management. During this project, I was actively involved in collecting and understanding the DataCo Supply Chain Dataset and performing complete data preprocessing using Power Query Editor. My responsibilities included correcting data types, handling missing values and duplicate records, removing unnecessary columns, renaming columns, creating derived columns such as Year and Quantity Range, and preparing the dataset for dashboard development.

I also worked on developing interactive Power BI dashboards by creating KPI cards, charts, maps, tables, slicers, and other visualizations to analyze sales, customers, products, shipping performance, discounts, markets, and profitability. I contributed to generating business insights from the dashboards, preparing the project report, presentation, GitHub repository, and GitLab submission. This internship strengthened my technical skills in Power BI, Power Query, DAX, data visualization, business analysis, and problem-solving, while also improving my confidence in working with real-world business datasets.

Virtual Internship 7.0

Team Member : Challa Jaya Deepika Reddy

The internship was a valuable opportunity to gain practical exposure to Supply Chain Analytics and Business Intelligence through collaborative project work. I contributed to the project by participating in dataset understanding, reviewing the preprocessing results, validating the cleaned data, and supporting the development and organization of Power BI dashboard visualizations. I also assisted in reviewing dashboard outputs, documenting business insights, preparing presentation content, and organizing the final project deliverables.

Working as a team helped me improve my communication, coordination, teamwork, and presentation skills while understanding how supply chain data can be transformed into meaningful business insights. This internship enhanced my knowledge of Power BI dashboards, data interpretation, report preparation, and collaborative project management, providing valuable experience in handling an industry-oriented analytics project.

10.Conclusion :

The Supply Chain Visibility System with Optimization Analytics project was successfully completed as part of the Infosys Springboard Virtual Internship 7.0 using the DataCo Supply Chain Dataset and Microsoft Power BI. This project transformed a large and complex supply chain dataset into meaningful business intelligence dashboards through a systematic process of data collection, data understanding, preprocessing, data cleaning, transformation, visualization, and business analysis. By applying Power Query Editor for preprocessing and Power BI for dashboard development, the project converted raw operational data into an interactive reporting solution that is easy to understand and analyze.

A total of seven interactive Power BI dashboards were developed, each focusing on a different aspect of supply chain operations. These dashboards analyzed overall business performance, product performance and pricing, customer behavior, geographic distribution, delivery and shipping efficiency, regional market performance, order and discount analysis, and profitability insights. The dashboards used KPI cards, charts, maps, treemaps, funnel charts, waterfall charts, tables, matrices, and slicers to present information in a clear and interactive format. The analysis showed that the company processed more than 180,000 orders, generated approximately 36.78 Million in sales, earned around 3.97 Million in profit, and maintained an average shipping time of 3.50 days while identifying Europe as the highest-performing market.

The project generated valuable business insights that can support data-driven decision-making in supply chain management. It helped identify top-performing products and departments, customer purchasing patterns, regional sales trends, shipping performance, delivery delays, discount impact, and profitability across different markets. The dashboards provide organizations with a transparent view of operational performance, helping managers monitor KPIs, identify performance gaps, improve logistics planning, optimize inventory management, and develop better pricing and customer strategies. The interactive nature of the dashboards also makes it easier to communicate business findings through reports and presentations.

Overall, this project provided practical experience in Business Intelligence, Data Analytics, Data Cleaning, Power Query Editor, DAX, Dashboard Development, and Data Visualization using Microsoft Power BI. It strengthened both technical and analytical skills while demonstrating the real-world application of data-driven reporting in supply chain operations. The project successfully bridges academic learning with industry practices and showcases how interactive dashboards can improve operational efficiency, business communication, strategic planning, and informed decision-making in modern supply chain management.

11. Acknowledgements :

We would like to express our sincere gratitude to the **Infosys Springboard Team** for organizing this highly enriching virtual internship program and for providing a structured, industry-aligned learning platform. The well-designed modules, milestone-based framework, and practical orientation of the internship enabled us to gain meaningful exposure to real-world business intelligence applications.

We extend our heartfelt appreciation to our mentor, **Ms. Nityasree**, for her consistent guidance, valuable feedback, and continuous encouragement throughout the project lifecycle. Her insights and structured direction played a significant role in shaping the analytical depth and professional quality of this project. We would also like to acknowledge our team members for their strong collaboration, accountability, and shared commitment toward achieving project objectives. The coordinated distribution of responsibilities, open communication, and collective problem-solving approach significantly contributed to the successful execution of this project. In addition, we extend our appreciation to the creators and maintainers of Microsoft Power BI documentation and community resources, which served as essential references during dashboard development and KPI implementation. We are also grateful to Kaggle for providing access to publicly available datasets that facilitated practical, hands-on analytical experience. The structured learning resources and practical exposure provided through this internship have significantly strengthened our analytical thinking, technical proficiency, and business interpretation skills. This experience has enhanced our ability to approach data with a strategic mindset, collaborate effectively within a team environment, and execute projects with professional discipline. Overall, this internship has been a meaningful milestone in our academic and professional journey,

reinforcing the value of data-driven decision-making and structured analytical execution in modern business environments.